Summary in Graph

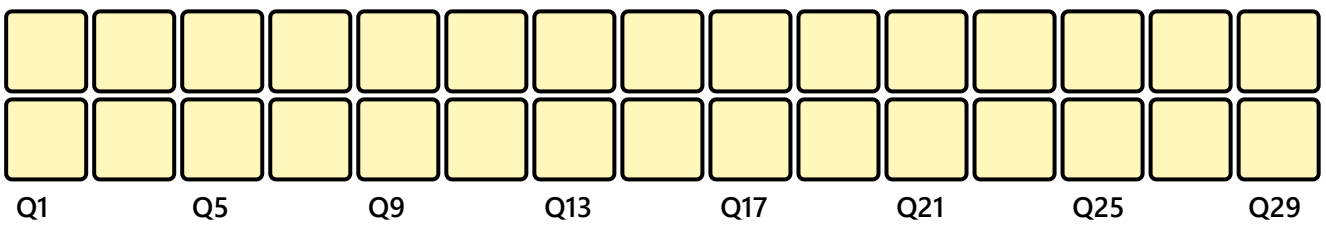
# Exam Summary (GATE Overflow CS Test Series 2026 | Linear Algebra | Shallow Test 1)

| | | | |
|---|---|---|---|
| Qs. Attempted: | 0 (0 + 0) | Correct Marks: | 0 (0 + 0) |
| Correct Attempts: | 0 (0 + 0) | Penalty Marks: | 0 (0 + 0) |
| Incorrect Attempts: | 0 (0 + 0) | Resultant Marks: | 0 (0 + 0) |

| | |
|---|---|
| Total Questions: | 30 (10 + 20) |
| Total Marks: | 50 (10 + 40) |
| Exam Duration: | 90 Minutes |
| Time Taken: | 0 Minutes |

## Individual Question Statistics

Select Metric: Marks Obtained




* Average Stats based on first-attempt stats of Top 10% (2) out of 18 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

## Technical

Q #1 | Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

Consider the matrix $B = [b_{ij}]$ defined for $1 \leq i, j \leq n$ by the formula $b_{ij} = i + 2j$ for all $i$ and $j$, where $n > 2$. The rank of matrix $B$ is _____.

Your Answer: | Correct Answer: 2:2 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #2** Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

Determine the rank of the following matrix:

$$B = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 1 & 0 & 2 \\ 2 & 3 & 8 \end{bmatrix}$$

Your Answer: | Correct Answer: 3:3 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #3** Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

The value of the Determinant D_4 is _____

$$D_4 = \begin{vmatrix} x & y & z & w \\ 2x & 2y & 2z & 2w \\ 3x+1 & 3y+1 & 3z+1 & 3w+1 \\ 4x+2 & 4y+2 & 4z+2 & 4w+2 \end{vmatrix}$$

Your Answer: | Correct Answer: 0:0 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #4** Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Given a matrix

$$G = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

if $G^T G = I$, then $\det(G)^3$ is:

A. 0
B. 1
C. 4
D. -1

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #5** Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

If $P = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 4 & 0 & 1 \end{bmatrix}$, the value of $\text{Adj } P$ is ____ (Mark all the appropriate choices) A. $\frac{P^{-1}}{\det P^{-1}}$ B. $\begin{bmatrix} -1 & 6 & -6 \\ 12 & -1 & -8 \\ -6 & 3 & 1 \end{bmatrix}$ C. $\det(P)P^{-1}$ D. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #6** Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Which of the following is/are correct? (Mark all the appropriate choices.)

A. The determinant of a triangular matrix is equal to the product of its diagonal entries.
B. The determinant of a matrix remains unchanged if a multiple of one row is added to another row.
C. The determinant of a block diagonal matrix is equal to the product of the determinants of its diagonal blocks.
D. The value of the determinant remains the same if a scalar multiple of one row is added to another row, provided no rows are swapped.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Let $X$ be an $n \times n$ invertible, orthogonal matrix (i.e., $X^T = X^{-1}$) such that $X^3 = X^T$. Define $Y = X^{-1}X^T$. Then $Y^2$ is equal to ____ (Mark all the appropriate choices).

A. $I$
B. $Y^{-1}$
C. $(Y^{-1})^T$
D. $X$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

If $B$ is an antisymmetric matrix of order $4$, what can be said about the matrix $B^8$?

A. It is a diagonal matrix.
B. It remains an antisymmetric matrix.
C. It transforms into a symmetric matrix.
D. None of the above.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Which of the following is/are correct? (Mark all the appropriate choices)

A. The sum of two skew-symmetric matrices yields a skew-symmetric matrix.
B. If $M$ and $N$ are two skew-symmetric matrices such that $MN = NM$, then the product $MN$ is skew-symmetric.
C. If matrix $M$ is skew-symmetric, then $M^k$ is also skew-symmetric for any positive integer $k$.
D. If $M$ is a non-singular skew-symmetric matrix, then $M^{-1}$ is skew-symmetric.

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Which of the following is/are correct? (Mark all the appropriate choices)

A. The product of two orthogonal matrices results in a matrix whose determinant is $1$.

B. The inverse of an orthogonal matrix is equal to its transpose, which implies $A^{-1} = A^T$.
C. The inverse of an orthogonal matrix is equal to its adjugate.
D. If $A$ is an orthogonal matrix, then $A^3$ is orthogonal.

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #11** Numerical Type | Award: 2 | Penalty: 0 | Linear Algebra

If $E = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is an involutory matrix, its index is _____.

Your Answer: | Correct Answer: 2:2 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #12** Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Let $\omega$ be a cube root of unity (such that $\omega^3 = 1$ and $\omega \neq 1$), and let $A = [a_{ij}]$ be an $3 \times 3$ matrix, where $a_{ij} = \omega^{i+j}$. Then, $A^k = I$ for some integer $k$ when $k$ is equal to _____.

A. 2
B. 3
C. 4
D. 6

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #13** Numerical Type | Award: 2 | Penalty: 0 | Linear Algebra

The determinant of the matrix

$$C = \begin{pmatrix} 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 1 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 4 & 0 & \frac{2}{5} & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & \frac{5}{2} & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 3 & \frac{1}{2} & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \end{pmatrix}$$

is _____

Your Answer: | Correct Answer: 1:1 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #14** Numerical Type | Award: 2 | Penalty: 0 | Linear Algebra

The value of

$$\begin{vmatrix} cosec^2\theta & \cot^2\theta & 1 \\ \cot^2\theta & cosec^2\theta & -1 \\ 36 & 34 & 2 \end{vmatrix}$$

$$+$$

$$\begin{vmatrix} \sin\alpha & \cos\alpha & \sin(\alpha+\delta) \\ \sin\beta & \cos\beta & \sin(\beta+\delta) \\ \sin\gamma & \cos\gamma & \sin(\gamma+\delta) \end{vmatrix}$$

equals to _____ .

Your Answer: | Correct Answer: 0:0 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #15** Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

If $a$, $b$, $c$ are positive and unequal, then the value of the determinant

$$|A| = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$

is _____.
(Mark all the appropriate choices)

A. Always Negative
B. Always positive
C. If a=1,b=2,c=3, $|A| = -18$
D. If a=1,b=2,c=3, $|A| = -3$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #16** Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

There are two values of $a$ which make the determinant

$$A = \begin{vmatrix} 1 & -2 & 5 \\ 2 & a & -1 \\ 0 & 4 & 2a \end{vmatrix}$$

equal to 86. The sum of these two values is

A. 4
B. 5
C. -4
D. 9

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #17** Numerical Type | Award: 2 | Penalty: 0 | Linear Algebra

Let $R$ be the following matrix:

$$R = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Find the determinant of $R$.

Your Answer: | Correct Answer: 1:1 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #18** Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

For the system of equations:

$$x + 2y + 3z = 1$$

$$2x + y + 3z = 2$$

$$5x + 5y + 9z = 4$$

Mark all the appropriate choices:

A. there is only one solution
B. there exist infinitely many solutions
C. there is no solution
D. $x = 0, y = -1, z = 1$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #19** Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

The following system of equations given below has:

$$2x + 3y - z = 0$$

$$x - y - 2z = 0$$

$$3x + y + 3z = 0$$

A. No Solution
B. Only Trivial Solution
C. Infinitely many non-trivial solutions
D. Exactly two solutions

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #20** Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Which of the following is/are correct? (Mark all the appropriate choices)

A. If a matrix $A$ has an eigenvalue $\lambda$ with corresponding eigenvector $x$, then for any scalar $c$, the matrix $A + cI$ (where $I$ is the identity matrix) has eigenvalue $\lambda + c$ corresponding to the same eigenvector $x$.
B. The determinant of a matrix $A$ is the product of its eigenvalues (counted with multiplicity).

C. If a matrix $A$ is diagonalizable, then the eigenvalues of $A$ are the entries on the diagonal of the diagonal matrix $D$ such that $A = PDP^{-1}$, where $P$ is the matrix of eigenvectors.

D. If two matrices $A$ and $B$ are similar (i.e., $A = PBP^{-1}$ for some invertible matrix $P$), then $A$ and $B$ have the same eigenvalues.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #21** Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Find the value of $x$ that makes the matrix $A$ Hermitian.

$$A = \begin{bmatrix} 3 & 2-i & 4 \\ 2+i & x & 5 \\ 4 & 5 & 7 \end{bmatrix}$$

A. $2+i$
B. $1-2i$
C. $2-i$
D. $1-2i^2$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #22** Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

The matrix $C = \begin{bmatrix} 5 & 5 & 5 & 5 & 5 \\ 5 & 5 & 5 & 5 & 5 \\ 5 & 5 & 5 & 5 & 5 \\ 5 & 5 & 5 & 5 & 5 \\ 5 & 5 & 5 & 5 & 5 \end{bmatrix}$ has eigenvalues that are _____

A. $0, 0, 0, 25, 0$
B. $0, 0, 0, 5, 20$
C. $5, 5, 5, 5, 5$
D. $10, 10, 10, 10, 10$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #23** Numerical Type | Award: 2 | Penalty: 0 | Linear Algebra

If the eigenvalues of a $2 \times 2$ matrix $B$ are $2$ and $5$, the sum of the eigenvalues of the matrices $(B+3I)^{-1}$ and $(B+7I)^2$ is _____.

Your Answer: | Correct Answer: 225.32:225.33 | Not Attempted | Time taken: 00min 03sec | Discuss

**Q #24** Numerical Type | Award: 2 | Penalty: 0 | Linear Algebra

Let $B$ be a real $2 \times 2$ matrix. If $4-3i$ is an eigenvalue of $B$, $\det(B) = \gamma$, and $\text{trace}(B) = \delta$, then the value of $3\gamma + 2\delta$ is _____.

Your Answer: | Correct Answer: 91:91 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #25** Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Given the matrix $B = \begin{pmatrix} 4 & 3 \\ 3 & 4 \end{pmatrix}$, which of the following are the eigenvectors of $B$? (Mark all the appropriate choices)

A. $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} -3 \\ 0 \end{pmatrix}$

B. $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} -3 \\ 3 \end{pmatrix}$

C. $\begin{pmatrix} -1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ -3 \end{pmatrix}$

D. $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 2 \end{pmatrix}$

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #26** Numerical Type | Award: 2 | Penalty: 0 | Linear Algebra

Let $A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 4 & 2 \\ 2 & 1 & 5 \end{bmatrix}$. For what value of $k$, the vector $\begin{bmatrix} k \\ 1 \\ 1 \end{bmatrix}$ is an eigenvector of matrix $A$? Then, calculate the value of $k^3 + 3k + 2$.

Your Answer: | Correct Answer: 2:2 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #27** Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

The matrix

$$B = \begin{bmatrix} 4 & 1 & 1 \\ 1 & 4 & 1 \\ 1 & 1 & 4 \end{bmatrix}$$

has three distinct eigenvalues, and one of its eigenvectors is

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Which one of the following can be another eigenvector of $B$?

A. $\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$

B. $\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

C. $\begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$

D. $\begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #28** Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Which of the following matrices have an LU decomposition? (Mark all the appropriate choices) A. $\begin{bmatrix} 2 & 3 & 5 \\ 6 & 13 & 23 \\ 4 & 10 & 19 \end{bmatrix}$ B. $\begin{bmatrix} 1 & 3 & 5 \\ 2 & 6 & 10 \\ 3 & 9 & 14 \end{bmatrix}$ C. $\begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$ D. $\begin{bmatrix} 2 & 3 & 1 \\ 4 & 6 & 3 \\ 1 & 2 & 0 \end{bmatrix}$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #29** Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

P: The matrix

$$A = \begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix}$$

has exactly one LU factorization, where L is a lower triangular matrix and U is an upper triangular matrix, diagonal of L(or U) consists of ones. Q: The matrix

$$B = \begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix}$$

has no LU factorization where L is a lower triangular matrix and U is an upper triangular matrix. For the given statements P and Q, A. P is true and Q is false B. Both P and Q are true C. P is false and Q is true D. Both P and Q are false

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #30** Numerical Type | Award: 2 | Penalty: 0 | Linear Algebra

For the matrix

$$M = \begin{bmatrix} 4 & 8 \\ 2 & 6 \end{bmatrix},$$

find the value of $u_{22}$ in its $LU$ decomposition, where $L$ is a lower triangular matrix with ones on the diagonal, and $U$ is an upper triangular matrix.

Your Answer: | Correct Answer: 2:2 | Not Attempted | Time taken: 00min 00sec | Discuss

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